Manual

Shopfloor Monitor (Graphic Machinery)

MDE-SFM 8.2

Version 1.0.23049

Last change on: 01.09.2020

Copyright

Contents

1 Overview Graphic Machinery

Purpose

The Graphic Machinery displays a company's machinery of a specific level (production hall, supervisor area, etc.). The Graphic Machinery displays the master data and also the actual data that is currently recorded as part of the Machine Data Collection (MDE) and Shop Floor Data Collection (BDE).

Implementation notes

You use the function package if:

- You want to display the available machines of your company and the relevant actual data.

Integration

The Graphic Machinery uses the current MDE data (e.g. machine status, current quantities) and the current BDE data (e.g. operations logged on). You can also display the current statuses of terminals or ZKS accesses (Access Control).

Features

- Graphic display
  - Graphic display of the machinery of a company: of a production hall, of a department, of a supervisor's area, etc.
- Two-level structure
  - The two-level structure includes separate machines and their assignment to machine groups, cost centers and lines.
- Display of the current machine statuses
  - The current machine status is visualized using different colors
- Tooltip
  - The info window (tooltip) provides detailed information
- Quantity progress
  - A bar chart visualizes the actually produced number of pieces and compares this number to the planned target number of pieces.
- Layout editor
  - Graphic editor to create layouts of halls where the symbols for machines, machine groups and production lines can be placed according to your requirements.

2 Graphic Machinery

Overview

Menu	Production facility management → Current information → Graphic machinery Information management → Production overview → Graphic machinery HR management → Access control → Security control center
Transaction code	mpark
Function authorization	mpark

Purpose

The graphic machinery provides a clear presentation of important workplace and production information and the relevant contexts in space and time. The user can create and display different presentations (so-called layouts) of all organizational units (e.g. halls, departments).

You can use the *Graphic machinery* as a quick overview tool for the production manager. But the application can also be used in the shop floor to provide a current overview of production statuses on large screens.

Integration

In the graphic machinery, you can visualize up-to-date status information of objects from various HYDRA products:

- BDE/MDE: Machines, groups, line groups, key performance indicators (KPI), logged in operations
- MW: Terminals
- WRM/EMG: Resources (no file-based resources, no DNC resources)
- MPL: Material buffers
- ZKS: Accesses, access groups

Requirements

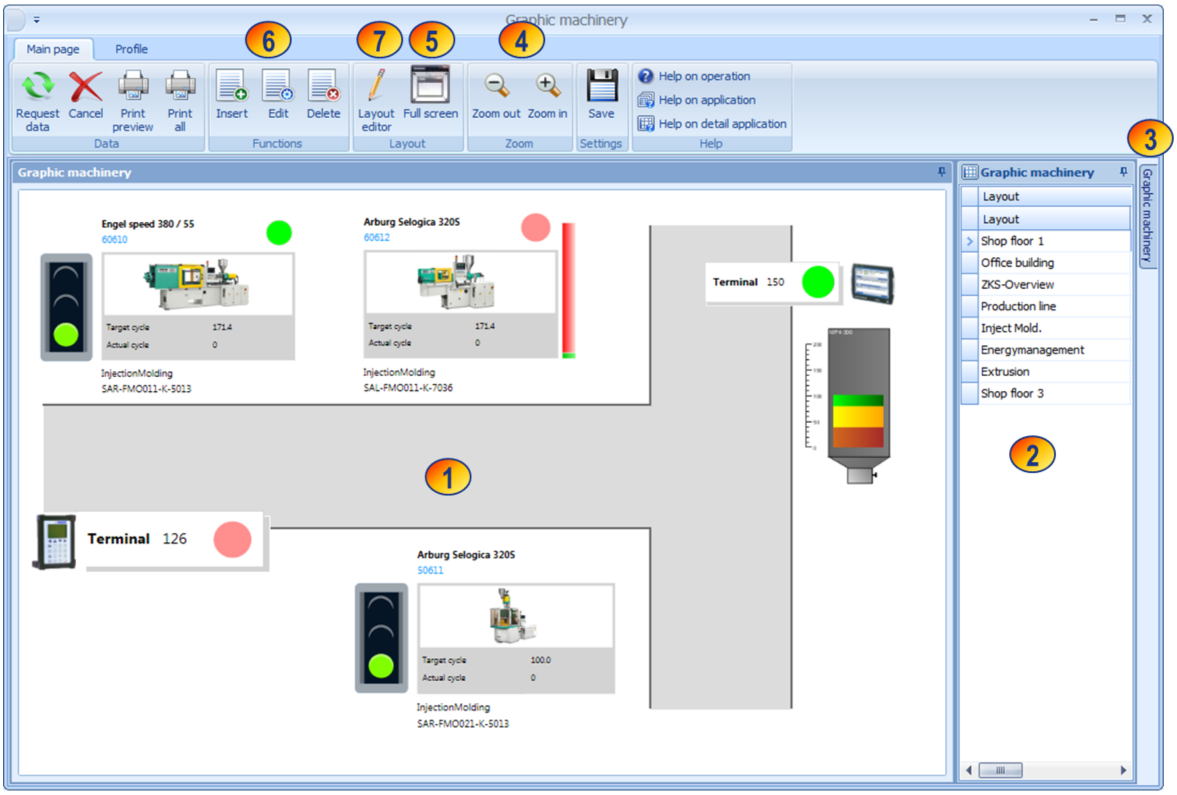
Set up the relevant objects you want to visualize on the Station Andon Board in the system master data.

Use the Layout editor to create the layouts required.

2.1.1.1 Structure of graphic machinery

You can use the *Graphic machinery* to show different layouts of different organizational units. You can generate these layouts according to your requirements.

To navigate through the layouts, use the table on the right hand side (2). The table contains the layout names and additional information on the different layouts. Left-click a table row to switch to the relevant layout which is then displayed in the layout panel (1).



(1) Layout panel

(2) List of available layouts

(3) Detail panel with information on the currently selected layout.

(4) Zooms in and out displayed layouts

(5) Starts the full screen mode; use the "Esc" button to cancel this mode.

(6) Editing functions to create a new layout and edit or delete an existing layout.

(7) Calls the the layout editor for graphic editing of the layout.

The application updates the statuses of machines, groups, terminals, etc. at regular intervals (every 30 seconds).

You can only request the *Help* function via F1 if the list of created layouts has the focus. In other cases, call up the *Help* function using the *Help on application* button.

Operation

Use the table on the right hand side of the screen (2) to navigate through the graphic machinery. Click a table row to switch to the requested layout.

Click the *Save settings* button to save the width of the table on the right hand side user-specifically. You cannot hide the table completely. Tip: You can "fold away" the table to the side using the respective button. You can find additional information on general operation in the documentation "Operation of the MES Operation Center".

If you right-click an icon in the layout panel (1), a context menu opens where you can switch to other applications, provided that you have the required authorization.

When you mouse over a template in a specific layout, a tooltip specific to each object opens showing additional information. You cannot configure the display time.

Details on the information displayed in a tooltip are described in the document MOC_GraphicMachinery2Edit.pdf, section *Notes on templates*.

Field descriptions

The following field descriptions refer to the table "Layout" and the detail view "Layout":

Layout

Layout name, e.g. "Hall 1"

Comment

Comment that describes the layout in more detail.

Sequence

If you specify a sequence/order and sort the table by the "order" column, the application always shows a particular layout by default when you call up the function. Please note that the defined order is not user-related.

Responsibility area

You can optionally assign a responsibility area to a layout. In this case, only users who are authorized for the entered responsibility area can use this layout.

Internal ID

The system assigns this number automatically every time you add a new layout. You cannot change this number.

File name

The layout configuration is saved in a file on the server. The system automatically assigns the file name when you add a new layout. You cannot change this file name.

Type

Always "N"

Modified by / Modified on

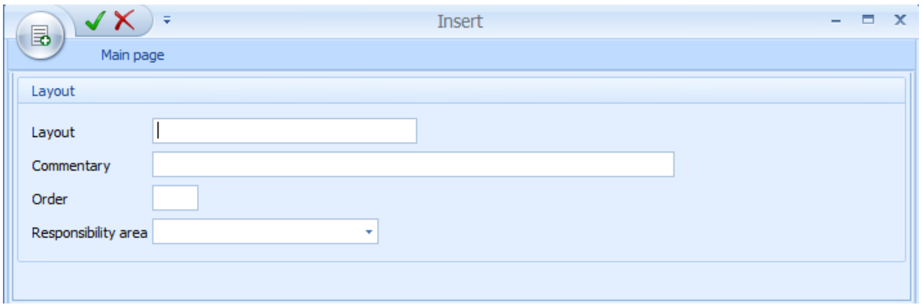
This information refers to the last time a layout entry of the table was edited, and not to the last time the graphic layout was edited.

Toolbar

 **Insert**

Function authorization: mparkle.create

Add a new entry to create a new layout. Please provide the following information: -

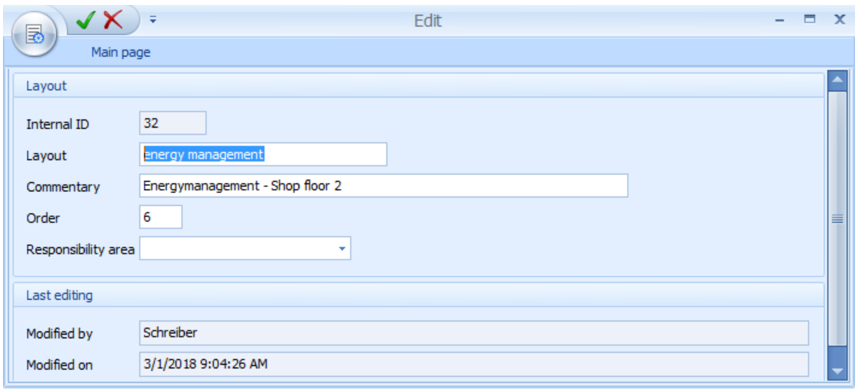


- Layout: Layout name
- Comment: Description
- Order: you can sort the list by this column
- Responsibility area (optional)

**Editing**

Function authorization: mparkle.edit

You may change the following data in an existing layout



- Layout
- Comment
- Order
- Responsibility area

Delete

Function authorization: mparkle.delete

The system deletes the layout entry from the database, once you have confirmed the deletion.

Please note: The file that saves the actual layout information on the server is not deleted.

Layout editor

Function authorization: mparkle.layedit

Calls up the Layout editor for the entry currently selected in the list.

Full screen

Zooms in the active layout to the full screen size. Press "Esc" to exit the full screen view.

Zoom out

Zooms out the current layout in the application.

Zoom in

Zooms in the current layout in the application.

Search

Use these two drop-down list boxes to search for specific objects in layouts. Select the object type (workplaces/machines, groups, line groups, material buffers, resources, terminals, accesses and access groups) in the upper field. Select the number of the searched object in the lower field.

Once you have selected the object, the list of available layouts indicates all layouts containing the relevant object. The application shows one of these layouts automatically. The searched object is highlighted with a red frame.

When you save a layout, the application identifies the objects included in the layout. You might have to save older layouts once more in order for the application to select and highlight the included objects.

Selection criteria

Selection criteria are not available in the graphic machinery. When you start the application, the application loads all configured layouts automatically and displays the first layout in the list.

If you specify the layout order and sort the list using the "order" column, the application always shows a particular layout by default when you call up the function. Please note that the defined order is not user-related.

2.1.1.2 *Additional features of the graphic machinery*

Opening full screen layouts

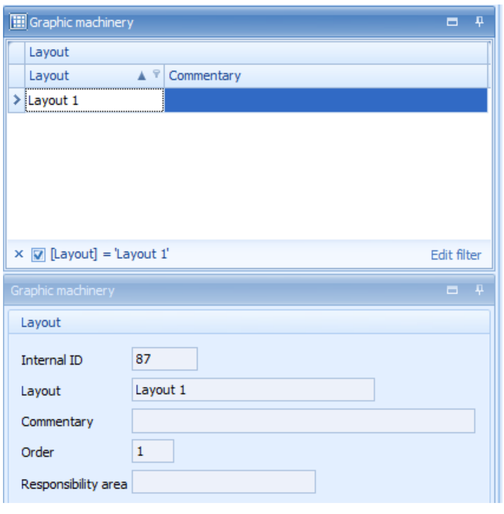
Use the menu editor to create an entry opening a defined layout in normal view or full-screen mode in the menu of a specific MOC client (menu depending on the user --> USER scope).

You can only use this function if the menu editor and the corresponding function authorization *mparklink* are available.

Proceed as follows if the menu editor is used:

1. Open the menu editor:
   Menu --> Extras --> Menu editor
2. Drag the graphic machinery to the required menu and change the values (enable advanced settings).
   - **Caption:** select individually, e.g. Mpark layout 1
   - **ID**: GraphicMachinery2
   - **Command**: mpark
   - **Parameter**: fullscreen id=<Internal ID of the layout>
     The parameter "fullscreen" opens the defined layout in full screen mode. The details pane of the graphic machinery shows the internal ID of the layout.

Example: “Layout 1” has the internal ID 87 (see screenshot).



Menu editor

If you use the "auto start" function, you can open a defined layout in full screen mode when starting the MOC.

Auto start: right click and select *Add to autostart*.

Setting the refresh interval

The refresh interval refers to the time interval between two update runs in the graphic machinery. You can set the interval via an INI configuration.

The value corresponds to the time interval in milliseconds between two update runs.

The default value of 60000 milliseconds (60 seconds) applies, if you enter a value less than 60000 milliseconds (60 seconds) or if nothing is entered.

Configuration:	
INI configuration name:	MPARK
Section:	MPARK2
Key:	REFRESHRATE
Value:	240000 (corresponds to 4 min)

Disabling offline messages

If the graphic machinery cannot establish a connection to the server (offline), the application displays an error message every time you request data.

You might not want this behavior, especially if you use the full screen mode. Configure the INI configuration as follows to prevent the error message from being displayed:

Configuration:	
INI configuration name:	MPARK
Section:	MPARK2
Key:	SUPPRESSCONNECTIONERRORMESSAGE
Value:	y Default value: n

If you start the graphic machinery the next time and you cannot request data (as a connection to the server cannot be established), the system only records entries in the MOC Log window.

But to inform the user, the window title of the full screen indicates the note "(Offline)" behind the layout name.

Automatic layout change

In the graphic machinery, you can change layouts automatically in the full-screen mode. You can configure the display duration and the order of changing the layouts.

Select the layouts and specify the display duration

Hold the CTRL key down to select several layouts at once in the table. Then click the button "configuration" in order to define the display duration.

A dialog opens where you can configure the display duration in seconds for a layout. The least interval for the automatic change is 30 seconds.

Once you have confirmed the dialog, the application stores the layouts and the display duration in a configuration file (LayoutRotation.config) in the MOC user directory.

Starting and stopping the layout change

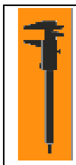
Click the button *Start* to start changing the layouts. After reading the stored settings, the system opens the full screen. Then the application changes layouts in full screen mode at the specified interval.

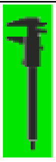
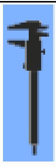
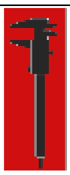
The time it takes to display the next layout depends on the number of objects displayed in the layout.

Click the button *Stop* or close full screen mode to stop changing the layouts automatically.

Objects "Inspection" and "Inspection points"

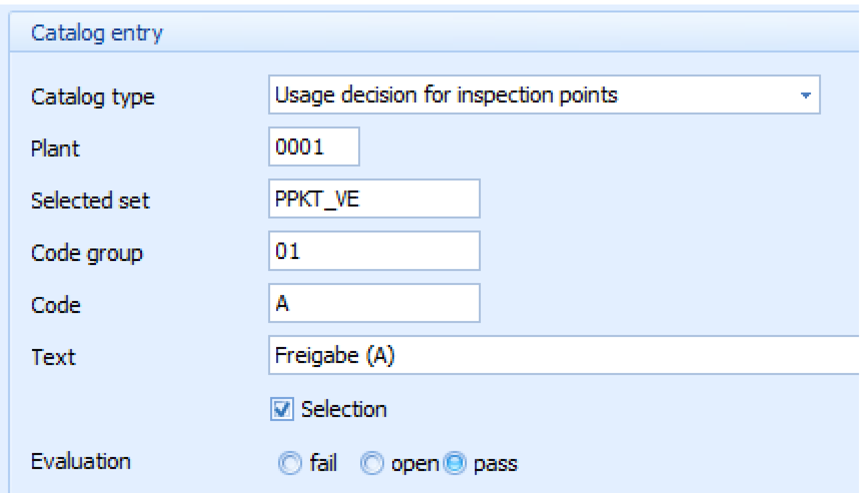
If the in-production inspection is activated and data is collected for inspection points, the current inspection status can also be integrated. To show the inspection status, select the object "inspection" (caliper symbol). The background color illustrates the different inspection statuses.

	Inspection status = due An operation including an inspection step has been logged on to the machine/workplace. At least one inspection point is available for this inspection step.

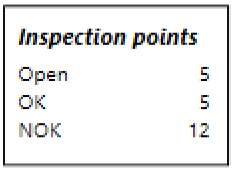
	Inspection status = checked An operation including an inspection step has been logged on to the machine/workplace. For this inspection step, no inspection point is available.
	Inspection status = inspection not possible The operation logged on to the machine/workplace does not include an operation step or no operation is logged on to the machine/workplace. Machines/workplaces of workplace type "F – in-production inspection" are inspection stations. To this type of machines/workplaces, no operations are logged on. These machines/workplaces always have a blue "inspection status".
	Inspection status = error If the system cannot identify any inspection step for a logged on operation because of an error, then the inspection status is displayed in red.

You can not only show the inspection status, but you can also show the number of open or closed inspection points for the logged on operation. Select the object "inspection points" to this end. With the number of closed inspection points, the system also displays the number of pass and fail inspection points. Condition: with the operation, an inspection step must have been logged on to the machine/workplace. The classification of an inspection point as pass or fail is based on the inspection point decision.

You define the inspection point decisions in the application *Catalog* of the quality management master data using the catalog type *Usage decision for inspection points*. The screenshot below illustrates the possible evaluations of an inspection point.

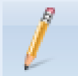


The screenshot below illustrates the entries for the object "inspection points".



3 Graphic Machinery - Layout Editor

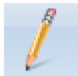
Overview

Menu	Production facility management → Current information → Graphic machinery Human resources management → Access control → Security control center Information management → Production overview → Graphic machinery
Transaction code	Mpark
Function authorization	mparkle.layedit
Function key	 Layout editor

Purpose

Use the layout editor of the Graphic machinery to "design" layouts showing user-defined organizational units (e.g. halls, departments, etc.).

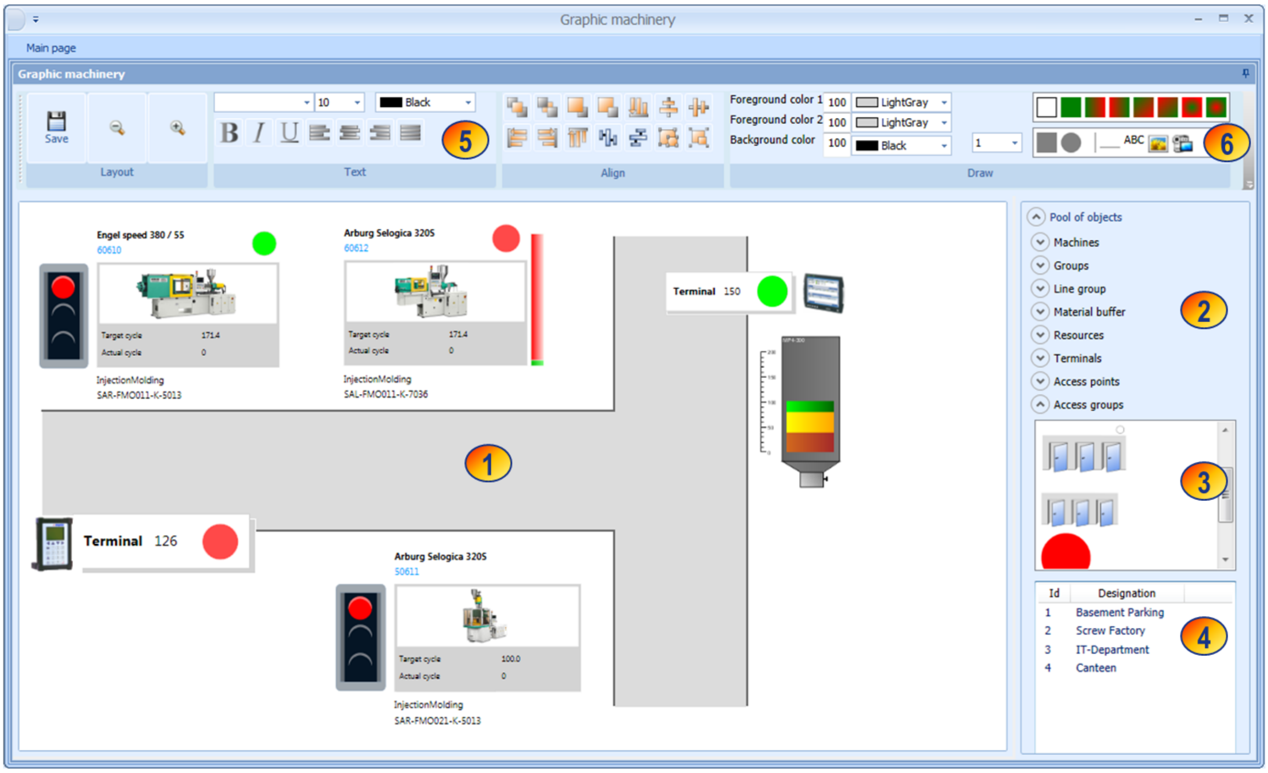
Integration

The layout editor is integrated in the application Graphic machinery. Use the button  to start the layout editor.

Requirements

The objects that you want to show in a layout must be available in the system master data. If necessary, you must create the objects in the master data. And you must have created a layout entry for the Graphic machinery.

3.1 Layout editor



(1) Layout panel

(2) Pool of objects: If you click an object type (e.g. machine), the dialog shows the following additional data:

(3) Templates of the object type

(4) Available objects (e.g. machines)

(5) The upper panel shows the "tools" you can use to edit a layout graphically (this is an example; chapter Toolbar describes all available icons).

(6) Graphic objects: templates, rectangle, circle, lines, text, image, video and/or video stream (mms).

In the layout editor you cannot start the help via F1. Start the help by clicking the button *Help on application*.

Toolbar

Please note that the layout editor does not provide a quick launch bar and you cannot minimize the toolbar.

The layout editor does not support functions of the MES Development Suite.

The layout editor does not support the functions "print preview" and "print all".

***Layout* category**

Save

Click "save" if you changed the layout.

Note: No confirmation prompt appears when exiting the application, even though changes have been made.

Zooms out the layout in the layout editor.

Note: Changing the size in the layout editor does not affect the data displayed in the graphic machinery.

Zooms in the layout in the layout editor.

Note: Changing the size in the layout editor does not affect the data displayed in the graphic machinery.

***Text* category**

You can use the functions described below to format a static text that you integrated in the layout using drag and drop via the button ABC.

Changes always affect the entire text within a text field.

Note:

- the buttons do not show current formatting of a text
  and
- font type, font size and color are also reset if you change the format (e.g. change to *italics).*

Arial

Changes the font of a static text.

10

Changes the font size of a static text.

Black

Changes the font color of a static text. Changes the entire text included in the text field.

Shows the text in bold letters.

Click the button once more to undo formatting.

Shows the text in italic letters.

Click the button once more to undo formatting.

Underlines the text.

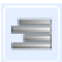
Click the button once more to undo formatting.

Left aligns static text:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epoqwieqoi

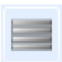
Center aligns static text:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epoqwieqoi



Right aligns static text:

asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epoqwieqoi

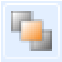


Justifies static text:

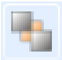
asasjaskj sjd aksjda ksjdlaksdj
alksjdalksjda lkdjklaksöa jdlaksjd
aksjdak lsdja iedkpo qwie qowiep
qoiweq poeiqow oqw pow iqpo
weipoq eipqowi epoqwieqoi

***Alignment* category**

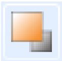
Use the functions of this category to align graphic objects.



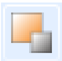
Brings the currently selected item to the foreground



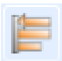
Brings the currently selected item into the background



Moves the currently selected item one level forward.



Moves the currently selected item one level back.



Left aligns the selected items.

Right aligns the selected items.

Top aligns the selected items.

Bottom aligns the selected items.

***Draw* category**

Use the functions described below to format a graphic object that you integrated in the layout using drag and drop via the buttons 

.

Changes always refer to all selected objects.Note:

- The buttons do not show current formatting of a graphic element and
- the color is also reset if the format is changed (e.g. changed color gradient direction).

Background color 100 LightGray

Changes the object color for objects without color gradient.

Changes the first color for objects with color gradient.

Changes the background color for text fields; you can change the foreground color by choosing a color in the *text* category.

2nd background color 100 LightGray

Changes the second color of objects with color gradient.

With text fields, the background color changes; you can change the foreground color by choosing the color in the *text* category.

Line color/border color 100 Black

Changes the line color.

ABC

Changes the border color of the object.

1

Changes the line thickness.

ABC

Specifies the fill effect, i.e. fill or gradient.

With text, the background is specified.

Left-click to select a fill effect before drawing one of the objects - ABC.

Click this icon if you want to draw a rectangle (rectangle, square). Drag and drop the item to the layout.

Click this icon if you want to draw a circle or oval. Drag and drop the item to the layout.

Click this icon if you want to draw a vertical line. Drag and drop the line in the defined thickness (see above) to the layout.

Once you have dropped the icon in the layout panel, a square is shown. You can change the size and the shape of the object as long as it is selected.

Note: You cannot draw diagonal lines.

Click this icon if you want to draw a horizontal line. Drag and drop the line in the defined thickness (see above) to the layout.

Once you have dropped the icon in the layout panel, a circle is shown. You can change the size and the shape of the object as long as it is selected.

Note: You cannot draw diagonal lines.

Click this icon if you want to enter a (static) text. Drag and drop the text in the defined font, font size, color and/or background color (see above) to the layout.

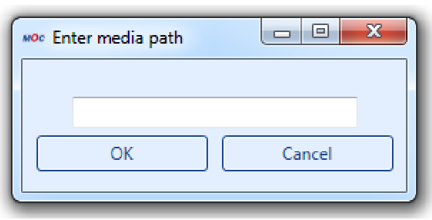
Once you have dropped the icon in the layout panel, the application displays a text field showing the letters "ABC". You can change this text as long as the object is selected.

Use this icon to insert an image into the layout. Once you have dropped the icon in the layout panel, a "File Open" dialog appears where you can select and insert the image into the layout.

The following formats are supported: jpg, png, bmp, gif.

Note: The image is transferred to the layout, i.e. to the zip file of the layout. Therefore, the file size affects the load time of the layout. We recommend using compressed file formats for images. The file size should range between double-digit and triple-digit kilobytes at most.

Use this icon to integrate videos or video streams. Select the icon, drag it into the layout and then drop it. A dialog opens where you can enter the path of the video or video stream.



The following formats are supported:

- Video: mp4, wmv
  You can enter an absolute or relative path referring to the MOC directory.
- Video stream: mms
  The MMS protocol (Microsoft Media Server Protocol) is a protocol developed by Microsoft that has been designed to transfer multimedia streams. The stream has to start with mms://, e.g. `mms://192.168.50.10:8080/`

Note: The video itself is not integrated in the layout, but the path to the video. Therefore, make sure that the video is stored in a central location that all MOC clients can access.

Note: If you select an object in a layout, the functions provided are adjusted to the properties of the selected object. Except for these options:

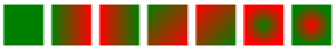
- Underlined
- Alignment (right, left, center, justify)

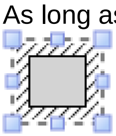
Operation

Note that the layout editor does not provide an "undo" function.

Integrating a graphic object, e.g. rectangle, line, text, image, etc. in the layout

In general, you can position a graphic object in the layout panel as follows:

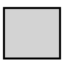
1. If you want to add a rectangle, circle or text, choose the fill effect at first by left clicking: . The selected icon is highlighted in color.
2. Left-click to select an object: 

. The selected icon is highlighted in color.
3. Hold the left mouse button down and drag this object to the layout panel and drop it at the required position (drag&drop).
4. If you insert an image or video/video stream, a dialog opens to select the image or an input field opens where you can enter the path of the video and/or video stream.
5. As long as the object is selected,

you can change its properties or move the object to another position by drag&drop.

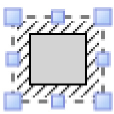
Selecting an object in the layout

To select a single graphic object, click that object. A "border" is shown around the object. Please find below an example for a square:

Before (before clicking):



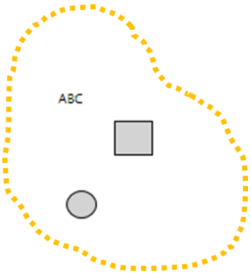
After (after clicking):



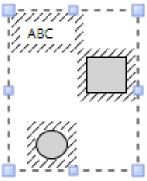
Selecting several objects in the layout

To select multiple objects at a time, hold the left mouse button down and "go" around the required objects so that an orange-colored, dotted circle occurs. Then release the left mouse button:

Before releasing the left mouse button:



After releasing the left mouse button:



Changes to the format, such as changing the (background) color now affect all selected objects.

Integrating objects, e.g. machine, group, etc. in the layout

Proceed as described below to display an object, e.g. a machine, group, in the layout:

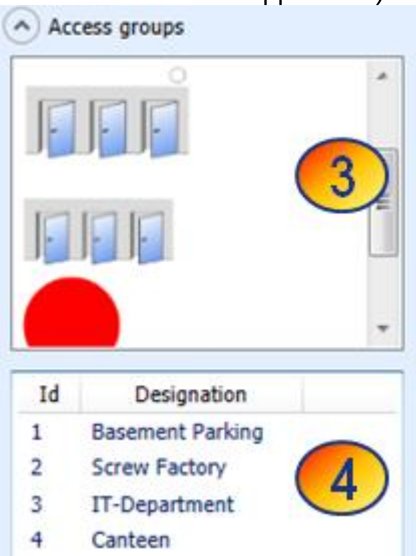
1. Make sure the right detail panel shows the object types. To do so, click the text "pool of objects". Then all object types are displayed (the number is only displayed in the above screenshot; the number is only for orientation and is not shown in the application):

2. Now click the object type you want to insert in the layout. Two windows appear below the object type:

   Top (3): the templates that are available for this object type

   Bottom (4): the objects of this type available in HYDRA

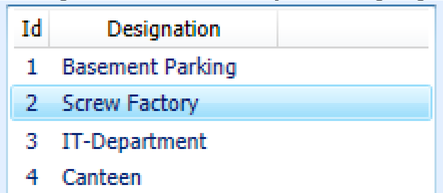
   (the numbers are only displayed in the above screenshot; the numbers are only for orientation and are not shown in the application):



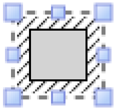
Please wait a few seconds until the below window shows IDs and designations. The number of columns and column headers in the below selection dialog depends on the object type.

3. Now select the template you want to show for the object in the layout by left clicking the template. The selected template is highlighted in color:

4. Now select the object you want to show in the layout by left clicking the object in the lower selection dialog. The selected object is highlighted in color:



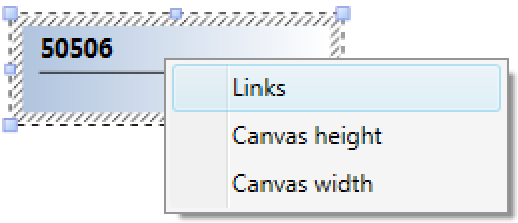
5. Hold the left mouse button down and drag this object from the selection window to the layout panel and drop it at the required position (drag&drop).
6. Click the object in the layout panel to change its size or to move it to another position by drag & drop.



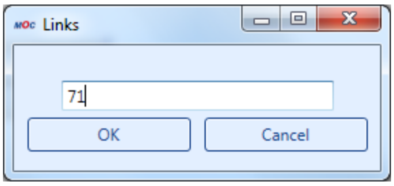
Inserting links between layouts

Use the context menu to add a link to another layout for objects. To do so, enter the appropriate layout ID. Proceed as follows:

1. Right-click the object you want to link. It does not matter if it is an inserted layout or another object.
2. Select the context menu item: Links



3. Enter the layout ID you want to add.



4. Confirm with ok.

<u>Note:</u> You can find the layout ID for the link in the list of layouts in the *Graphic machinery* (shop floor monitor). If the column indicating the layout ID is not displayed, add the column to the list by choosing the option "select columns".

Adjusting drawing canvas

Use the context menu of the layout editor to adjust the available drawing canvas. In the context menu, select the option "canvas height" and "canvas width" and enter the required values.

Templates in the Graphic Machinery

The graphic machinery provides ready-made templates for the below-mentioned object types.

- Workplaces/machines

- Groups (capacity groups)
- Lines (workplaces/machines of the "L" type)
- Material buffer
- Resources (no DNC and file resources)
- Terminals
- Access points
- Access groups
- KPIs of energy resources
- Machine-specific energy meters
- Key performance indicators (current, shift-related KPIs for workplaces/machines)
- Operations logged on (up to three operations)

You can change existing templates as part of a customization.

Deployment of layouts

You can pass created layouts from one system to the other. MPDV explains how to pass layouts as part of a service.

3.2 Notes on tooltip templates

Workplaces/machines

The tooltip shows the following data:

Workplace and machine data:

- Workplace – Unique identification of the workplace/machine according to the Configuration.
- Designation – Name of the workplace according to the Configuration.
- Workplace category – Workplace category according to the Configuration.
- Status – Status text of the status currently set for the workplace
- Status since – Beginning of the current status
- Target cycle – Current target cycle (format: hours:minutes:seconds per 1000 cycles)
- Actual cycle – Current actual cycle (format: hours:minutes:seconds per 1000 cycles)
- Yield (P) – Yield recorded for the workplace in the current shift (primary quantity unit)
- Scrap (P) – Scrap recorded for the workplace in the current shift (primary quantity unit)

Operation related data for the currently logged on operation. If several operations are logged on to the workplace at the same time, the data of the operation last logged on is displayed.

- MES order number
- OP designation

- Article – Article defined in the operation
- Article designation – Article designation defined in the order
- Tool – Tool defined for the operation
- Yield (P) – Yield posted for the operation (primary quantity unit)
- Scrap (P) – Scrap posted for the operation (primary quantity unit)
- Open quantity (P) – Open quantity posted for the operation (primary quantity unit)
- Rework (P) – Rework quantity posted for the operation (primary quantity unit)

Groups

The tooltip shows the following data:

- Group – Identifies the group according to the Configuration
- Designation – Name of the group according to the Configuration
- Cost center – Cost center of the group according to the Configuration

Line group

The tooltip shows the following data for line workplaces (type L):

- Workplace – Unique identification of the workplace/machine according to the Configuration.
- Designation – Name of the workplace according to the Configuration.
- Workplace category – Workplace category according to the Configuration.

Terminal

Terminal templates do not provide any tooltips.

Material buffer

The tooltip shows the following data:

- Material buffer – Identifies the material buffer according to the Configuration
- Material – Material number according to the Configuration

Resource

The templates for resources show the resource status.

- Resource – Identifies the resource according to the Configuration
- Resource type – Resource type of the resource according to the Configuration
- Designation – Name of the resource according to the Configuration
- Current storage location – Current storage location of the resource
- Resource status – Number of the current resource status

- Resource status – Name of the current resource status according to the Configuration

Access points

The tooltip shows the following data:

- Access point – Number of the access point according to the Configuration
- Designation – Name of the access point according to the Configuration

Access groups

The tooltip shows the following data:

- Access group – Number of the access group according to the Configuration
- Designation – Name of the access group according to the Configuration
- Location – Location of the access group according to the Configuration
- Number status OK – Number of access points with status OK
- Number status error – Number of access points with status "error"
- Number status not active – Number of access points with status "not active"
- Status color access group – for internal use

KPIs of energy resources

The templates for KPIs of energy resources do not provide any tooltips.

Machine-specific energy meters

The templates for machine-specific energy meters do not provide any tooltips.

Key performance indicators (current, shift-related KPIs for workplaces/machines)

You can choose from the following key performance indicators:

KPI	Formula ID used for changes in the formula management and for the configuration of limit values (optional)
Rate of capacity utilization	rcu
Assignment utilization rate	ocu
Techn. efficiency	tec_ef
Rate	yie_ra
Scrap rate	scr_ra
OEE	oee

KPI	Formula ID used for changes in the formula management and for the configuration of limit values (optional)
Availability	avail
Performance	pf_rat
Quality	qual
Machine run time	mch_rt
Actual utilization	act_ut
Yield utilization	yie_ut

Refer to the following document MDE_KPI_Configuration.pdf for further information on the configuration.

The templates for KPIs do not provide any tooltips. Create a static text and enter the workplace. This way, later on you will know the workplace this KPI refers to.

Operations logged on

The template shows up to three logged on operations including the following information on the operation:

- Article number
- Target quantity (P)
- Yield (P)
- Scrap (P)
- Scrap ratio in %
  The system calculates the relation of scrap (P) to the total yield quantity (P) and scrap quantity (P). The result includes two decimal places. The application indicates "undef." if the scrap rate cannot be calculated.

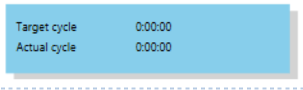
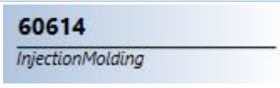
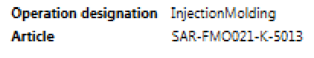
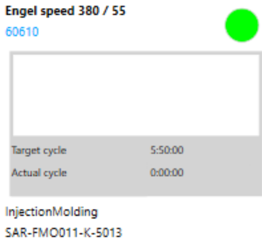
The values are displayed left aligned. The point in time of the operation logon specifies the order in which the application shows the data of the logged on operations. The operation logged on last is displayed first.

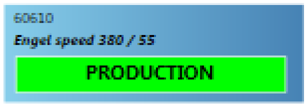
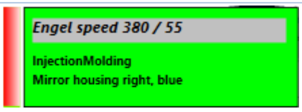
The template does not provide tooltips. Create a static text where you can enter the workplace. Therefore, you can identify later to which workplace this information refers.

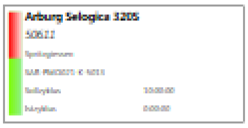
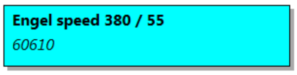
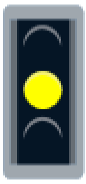
3.3 Description of the templates for workplaces/machines

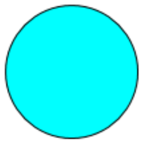
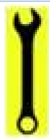
☼ Available as of service pack 13/2018:
In the Status assignment, you can specify a time for each machine status in field "Warning in

the Graphic Machinery". If a machine status lasts longer than the specified time, the machine status symbol starts to flash. If the template supports the flashing status symbol, the machine status is marked with ☼.

Template name	Screenshot	Description and content of the template
Machines_Cycles		Display of target and actual cycle for the workplace/machine. For the display, the format is used that is defined for the display in the MOC.
Machines_Description		Display of the workplace/machine number and of the name.
Machines_OperationArticle		Display of OP name and of article of the logged on operation. If several operations are logged on, the operation logged on last is displayed.
Machines_QuantityLevel		Graphic comparison of the target quantity and the yield produced up to now (in primary quantity unit) of the currently logged on operation. If several operations are logged on, the information of the operation logged on last is displayed.
Machines_Template1		Display of the following information: - Machine name - Machine number - Graphic display of machine status ☼ - Current target cycle and actual cycle of the machine in the format that has been defined for the display in the MOC. - Operation designation - Article name of the operation

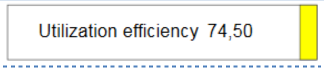
Template name	Screenshot	Description and content of the template
		You can use the free space to add an image of the respective machine.
Machines_Template2		Display of the following information: - Machine number - Machine name - Graphic and text display of machine status ☼ - Primary target quantity, yield and scrap of the operation logged on. - Current target cycle and actual cycle of the machine in the format that has been defined for the display in the MOC. - Operation designation - Article name of the operation - Graphic comparison of the target quantity and the yield produced up to now (primary quantity unit) of the currently logged on operation.
Machines_Template3		Display of the following information: - Machine number - Machine name - Graphic and text display of machine status ☼
Machines_Template4		Display of the following information: - Machine name - Operation and article name of the logged on operation - Color display of machine status as background color ☼

Template name	Screenshot	Description and content of the template
		- Graphic comparison of the target quantity and the yield produced up to now (primary quantity unit) of the currently logged on operation.
Machines_Template5		Display of the following information: - Machine name - Machine number - Operation and article name of the logged on operation - Current target and actual cycle of the machine using the unit that has been defined for the display in the MOC. - Graphic comparison of the target quantity and the yield produced up to now (primary quantity unit) of the currently logged on operation.
Machines_Template6		Display of the following information: - Machine number - Machine name - Color display of machine status as background color ☼
Machines_TrafficLight		Traffic light display including the following information ☼: - Color red: Status "Not assigned" - Color yellow: All statuses except "Production" - Color green: Status "Production"

Template name	Screenshot	Description and content of the template
Machines_StatusCircle		Color display of machine status ☼ using the colors defined with the status text.
Machines_MaintenanceStatus1 Available as of service pack 13/2018		If the activity/maintenance calendar includes at least one maintenance that is due, this maintenance due is displayed using a symbol. If there is at least one "red" maintenance due, the background color is red. If there is at least one "yellow" maintenance due, the background color is yellow. If there is at least one "blue" maintenance due, the background color is blue. If no maintenance is due, no symbol and no background color is displayed.
Machines_MaintenanceStatus2 Available as of service pack 13/2018		See Machines_MaintenanceStatus1

3.4 Description of the templates for KPIs

Template name	Screenshot	Description and content of the template
keyfigure_template_label		Display of the following information: - KPI name

Template name	Screenshot	Description and content of the template
		- current shift-related KPIs - color display of the KPI as background color
keyfigure_template_label_border		Display of the following information: - KPI name - current shift-related KPIs - color display of the KPI as bar
keyfigure_template_simple		Display of the following information: - current shift-related KPIs - color display of the KPI as background color